BHIMSHILA

PROJECT REPORT [Phase-I/II/III]

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CERTIFICATE

This is to certify that the work embodied in this report entitled **"BHIMSHILA"** has been satisfactorily completed by **MRIDU (24U021030) and SHRISHTI MISHRA (24U021029).** It is a bonafide piece of work, carried out under our guidance in the **Department of Computer Science and Engineering, Indian Institute of Information Technology, Bhopal** for the partial fulfillment of the Bachelor of Engineering during the academic year 2026 - 27.

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DECLARATION

We hereby declare that the following major project synopsis entitled “BHIMSHILA” presented in the is the partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering**. It is an authentic documentation of our original work carried out under the able guidance of **Dr. Sonal Telang Chandel**. The work has been carried out entirely at the Indian Institute of Information Technology, Bhopal. The project work presented has not been submitted in part or whole to award of any degree or professional diploma in any other institute or organization.

We, with this, declare that the facts mentioned above are true to the best of our knowledge. In case of any unlikely discrepancy that may occur, we will be the ones to take responsibility.

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AREA OF WORK

Our project, BhimShila, is targeted at developing an integrated disaster management platform using web technologies, data integration and AI assisted analysis. The system aims at allowing citizens, volunteers and coordinators to report emergencies and coordinate response activities via a central platform.

Disaster response typically requires information from a variety of sources, including weather forecasts, earthquake records, government alerts and geographical information. If information is collected and interpreted separately it may be difficult to get a clear picture of an evolving situation. BhimShila solves this problem by bringing in relevant external data sources and presenting the information in one system to support situational awareness and response planning.

The project involves development of web-based interfaces, backend services, database management and data-processing workflows. This includes deterministic risk scoring, AI-assisted incident-report processing, duplicate report identification, and incident triage to help organise incoming information. The platform also supports workflows for emergency reporting, mission coordination, resource management and alert communication. We reserve human confirmation for some public-facing actions, rather than relying completely on automated decisions.

We are building the system using Next.js, React, TypeScript and Tailwind CSS. The system uses a SQLite compatible database to persistently store data. It has external integrations like government disaster alerts, weather information, earthquake data and geographical services. The project also investigates offline support for specific reporting and coordination workflows, which allows for continuity in situations with unreliable network connectivity.

Through this work the project wants to show how software engineering, data integration and AI assisted information processing can be combined to improve the management of disaster related information and to support a more organised emergency response.

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ABSTRACT

A disaster management system must include ways to verify, interpret, distribute and link information to response teams. Information can be collected via many different platforms, there may be several reports of the same incident and internet coverage may be limited. The system described below is an example of a platform that attempts to unify this process.

The platform can be used to provide official alerts, weather, geospatial data, citizen SOS reports, volunteer updates and a coordinator dashboard. The regional intelligence module collects data from public sources such as NDMA SACHET alerts, Open-Meteo weather data, USGS earthquake data, news feeds and OpenStreetMap features. The risk scoring module works deterministically to calculate the risk score by looking at structured environmental data and also provides insight into what contributes towards the risk. Using AI, reports can be turned into structured data. However, the AI will always follow safety rules to ensure important rescue situations aren't downgraded. If reports match the same category, they can be grouped together based on proximity.

There are three user types within the system: citizen, volunteer and coordinator. Citizens can send SOS reports and will receive alerts or information on nearby shelters. Volunteers can complete tasks and send updates. Coordinators can look at the common view of an incident, verify signals, approve public actions, allocate teams and monitor the deployment. There is also a local queue option for selected actions within the field. Confirmation will be needed to complete actions and the system is designed to help humans make decisions.

The system follows the Next.js framework with React components written in Typescript. APIs are used along with a libsql/sqlite compatible data store to hold the data. The repo also includes deterministic tests for the main logic and integration tests for selected live data sources. This report will discuss the architecture, data flow, algorithms, implementation, any limitations and improvements. BhimShila is a prototype decision support system and is designed to help and not replace official warning systems, emergency services or verification on the ground.

Keywords: Disaster management, Early Warning Systems, Incident Triage, Geospatial Data, Voice, Offline First, Relief Coordination, Human In The Loop AI

INTRODUCTION

1.1 Background

A disaster such as flooding, earthquakes, cyclones and landslides can cause serious damage to human lives, infrastructure and the environment. To manage a disaster effectively, you need timely information, reliable communication and correct resource allocation. Response teams should also have good coordination. Collecting, analysing and communicating data is incredibly important to improve preparedness and organise emergency response activities. [1][2]

There are various sources of information available in India relating to disasters. These can include official alert systems, weather services, earthquake monitoring systems, geographical data etc. However these are separated and it will be hard to find all the information you need in one place. Interpretation and coordination will need to be done on emergency reports, official alerts and updates from the field. Therefore, a digital system is needed to help manage the information.

1.2 Problem Context and Motivation

Processes involved in responding to a disaster include identifying hazards, collecting incident reports, verifying information, prioritising emergency needs, allocating response teams and distributing updates. Having separate systems to manage this data can cause issues with scattered information, duplicate reports, missing or delayed reports of important incidents and difficulty tracking relief efforts. [3][4]

It can be difficult to determine which reports to focus on. Some incidents will require immediate emergency support but others will not be as urgent. Coordinators need to be able to verify reports properly and identify serious incidents. Volunteers need to have specific tasks and be able to give updates from the field. Citizens should be able to easily report incidents.

Internet coverage may not be great in the field. If the applications require constant internet, users may struggle to put in or send information. These are some of the issues that have driven the creation of BhimShila. The system is designed to collect disaster related information, incident reports and response coordination into one system.

1.3 Introduction to BhimShila

This is a web based system to provide disaster intelligence and emergency response coordination. It collects external disaster information, citizen reports and volunteer updates. Users will have access to a centralised system.

The system collects data from various sources including NDMA SACHET alerts, Open-Meteo weather, USGS earthquake, OpenStreetMap geographical data and selected news feeds. It contains data on official alerts, environmental conditions, seismic activity and surrounding

geographical information. This data will be processed to help users understand what is happening. [5]-[9]

There are three user types within the system: Citizen, Volunteer and Coordinator. Citizens can report SOSs and receive alerts or information on nearby shelters. Volunteers can manage their missions and send updates from the field. Coordinators can view incidents, analyse risk data, verify reports, allocate teams and monitor response activities.

There is also deterministic risk scoring and AI supported incident processing. Risk scoring looks at structured data to help provide more context on the risk. AI supported natural language processing helps convert spoken/typed information into structured incident data. To ensure important incidents aren't downgraded automatically, safety rules are put in place. Duplicate reports can be grouped together based on matching categories and defined geographical/time parameters.

1.4 Scope and Significance

The system will collect external alerts, environmental/geographical data, SOS reports, incident triage, duplicate reports, verification processes, mission allocation, volunteer updates and coordinator led response management. Some actions can be completed from a queue whilst offline and will be synced once connected.

The project includes front end development, back end services, data management, external API integration, data processing and AI support. The application has been developed using Next.js, React, TypeScript and Tailwind CSS. The data is stored using a SQLite compatible database.

The main feature of this system is the ability to bring together disaster related information and coordination. The system aims to help users organise reports, understand the risk indicators and coordinate response activities. Some public facing actions will require human confirmation.

This system is designed to be a decision support and coordination system and is not an official forecasting system for disasters. The risk scores provided are additional information to help prioritise and should not be considered as calibrated probabilities. This system is not designed to replace existing official warning systems. [1][5]

1.5 Report Organisation

The rest of this report will be organised as follows. Chapter 2 will look at the literature and current methods for disaster monitoring and emergency coordination. Chapter 3 will define the problem and objectives of the project. Chapter 4 will discuss the methodology and system flow. Chapter 5 will discuss the main algorithms used. Chapter 6 will look at the system architecture, flowcharts and data flow diagrams. Chapter 7 will discuss the tools and technologies used. Chapter 8 will show the results and analysis. Chapter 9 will conclude the report and future scope.

LITERATURE REVIEW

2.1 Disaster Management Information Systems

Disaster management involves preparation, early warning systems, emergency response, relief distribution and recovery. To manage disasters effectively, you need quick access to information, reliable communication and good coordination between citizens, volunteers and emergency services. Information and communications technology have played an important role in helping to collect, process and distribute this data.

A literature review was conducted to look at information systems relating to emergency management [3]. Research into knowledge management systems for disaster management [4] also highlights the importance of organising information and sharing knowledge between stakeholders. This shows that digital systems are important in emergency planning.

However, information can be split between different independent platforms, communication channels and manual data entry. This can make it difficult to gain an overall picture of incidents. This issue can be overcome by using information systems that collect the relevant data and processes.

This is what BhimShila does! It brings together external disaster information, citizen reports, incident review and response coordination. It is designed to help organise information and emergency related activities and is not meant to replace existing disaster management authorities.

2.2 Existing Disaster Alert Systems

Early warning systems help provide information on potential risks so that people and authorities can take appropriate action. However this depends on hazards being identified and warnings being made available in a timely manner and in an easy-to-understand format.

The SACHET portal in India is managed by the National Disaster Management Authority (NDMA) and helps distribute disaster alerts using the Common Alerting Protocol (CAP). [3] Using standardised alert formats allows compatible systems to share information. Targeting information to specific geographical areas and having multilingual options means the right people can receive warnings. [5][7]

Official alerts provide an important source of information however further steps may be needed. This can include reviewing local incident reports, identifying effected locations, allocating teams and noting updates from the field.

BhimShila integrates disaster related alerts. It aims to provide relevant information from external sources alongside incident reports and response coordination. This system is not a replacement for NDMA, India Meteorological Department (IMD) or any other official bodies. Official warnings will always be attributed to the relevant issuing body.

2.3 GIS and Geographical Data

Geographical Information Systems (GIS) can be used to help with disaster management. This helps link incidents and hazards to locations. Maps and geographical data can be used to show roadways, settlements, waterways, boundaries and other important factors for emergency planning.

A literature review was carried out to look at using web based GIS for natural hazard management [8]. This shows how using online geographical tools can help provide information on hazards. Web based mapping can be useful if multiple users need to view location based information without installing specific desktop software.

Using OpenStreetMap can provide valuable geographical data to include in applications. This can help to provide information on roadways, waterways, settlements and nearby facilities. This along with incident locations and environmental information can help users understand the geographical context of an emergency. [9] BhimShila uses geographical services to show incident related information and also pull in location based information from external sources.

An important part of disaster management is risk. This can be dependant on hazard conditions, exposure and local vulnerability. A digital system can use structured data to help prioritise incidents however the quality of data and scoring method will affect how useful this is. BhimShila uses deterministic risk scoring to add to the prioritisation information. This is not an AI generated score to help predict if a disaster will happen. [10]

GIS can help provide situational awareness however maps cannot be used to verify if an incident is real or if a response team has had action. This means geographical data should be integrated with incident verification.

2.4 Crowdsourcing and Crisis Informatics

During an emergency situation, there may be useful information provided by citizens, volunteers, local organisations, news sources and official bodies. Local people in the area can provide information on what the local conditions are like.

Crisis informatics looks at how information and communications technology can help provide information during disasters. Research into humanitarian information processing shows how text based information and public communications can help identify needs. [11][12]

However, crowdsourced reports can be incomplete, incorrect, outdated and duplicated. Different people may use different wording to describe the same event which can make it difficult to classify manually during large emergencies. An incident reporting system should help organise reports.

BhimShila Provides Citizen SOS Reporting and Structured Incident Review. Duplicate reports can be grouped together based on incident category, geographical distance and time. This will help reduce duplicate review. However, just because reports are close by and in the same time frame does not mean they are about the same incident.

Separating the reporting of information from the approval of specific public actions is important. A report from a citizen or automated interpretation shouldn't be automatically considered as an officially verified warning.

2.5 Artificial Intelligence and Natural Language Processing

Artificial Intelligence (AI) and Natural Language Processing (NLP) can help organise information that would be difficult to process manually. This may help with text classification, information extraction, summarisation and converting unstructured reports into structured data.

A literature review was carried out to look at using NLP for hydrometeorological hazard assessment [13]. This shows how text based information can be used to compliment traditional monitoring data. Rocca et al. looked into using NLP for humanitarian action. This showed how useful information can be extracted from humanitarian documentation and communications. More recent research has looked into using large language models for detecting, analysing and responding to disasters.

However, standard AI systems can give incorrect interpretations, miss key points and provide incorrect information. This can be critical in an emergency situation and incorrect prioritisation may affect decisions. Therefore, AI output should be used with the correct precautions.

BhimShila uses AI driven incident processing to help convert user descriptions into structured data. This will make reports easier to review and will reduce some manual processing. However, the severity of an incident isn't just determined by AI. Deterministic rules will ensure critical rescue situations have a minimum urgency. Automated processing won't down grade them.

This means safety critical rules have been separated from natural language processing. Predictable behaviour can be achieved using structured logic. Specific public actions have been left to be reviewed by a human. This acknowledges that whilst AI can help with decision making, it shouldn't be used to replace established authorities.

2.6 Emergency Coordination and Resource Allocation

Disaster response will involve collaboration between citizens, volunteers, coordinators and emergency organisations. Each group will have a different role. Citizens will report emergencies, volunteers will participate in field activities and coordinators will review information.

Research into knowledge management [4] highlights how important it is to share information between stakeholders. Incident reports and updates can be disjointed and make it hard to know the current status of an incident.

A centralised system can help with this by having structured incident reports, allocating missions, updating status and providing a shared view of operations. Incidents may also be prioritised based on urgency and available information to help deploy resources. However, the actual allocation may depend on team availability, accessibility, equipment needs and human judgement.

BhimShila provides processes for allocating missions, volunteer updates, incident review and monitoring by coordinators. There are three main user types: Citizen, Volunteer and Coordinator. This links reporting an emergency to reviewing and responding.

The system has been designed to provide defined coordination processes. This isn't a full inter-agency emergency management system. Good coordination between multiple organisations will depend on reliable communication, agreed processes, clear roles and correct permissions.

2.7 Offline Access and Reliability

During a disaster, communication infrastructure can be crippled. Field staff may need to record information however there may be little or no internet. This can limit reporting and coordination if the application requires constant internet for each process. Some offline functionality can be provided using Progressive Web Application (PWA) techniques however this depends on having suitable data storage and synchronisation. [15]

BhimShila will support the use of offline capabilities in specific scenarios such as SOS, transitions and field reporting depending on the implementation of the particular workflow. The data may be stored locally and sync once connectivity is available.

However there are problems involved with using offline features. Any update delays can occur along with many people making changes to the data whilst disconnected. The application must understand what data has been stored, what data needs syncing and if the data has been successfully added to the server. Offline features are not a guarantee that all functions will be available but will be reliable for specific activities.

2.8 Comparative Analysis of Existing Approaches

The approaches reviewed look at different aspects of disaster management. From official alert systems to provide alerts, GIS providing location data, crowdsourcing to add local data and AI to help process unstructured data. These features are connected through coordination workflows.

Table 1: Existing Disaster Management Systems Comparison

Approach	Contribution	Remaining Requirement	Relevance to BhimShila
Emergency management information systems [1, 2]	Information organization and decision support	Linking information to operational processes	Centralised incident handling
Official warning systems [3]	Disseminate standardised cautions	Mission tracking and local incident reports	Integration of external alarms
Web GIS [4]	Show information according to location	Check and follow up on location-based information	Location information for events
Crowdsourcing journalism [5]	Local observations	Duplicate and unverified report management	Incident review and SOS reporting
AI and natural language processing [5]–[7]	Extracts and organises data from text	Handling uncertainty and wrong outputs	AI-assisted processing with safety rules
Apps that work offline	Support for certain tasks during network outages	Synchronisation and update consistency	Selected offline reporting workflows

It showcases how the technologies can be complementary. It can offer a single platform where selected functionality can be provided, however to ensure a reliable system data integrity, validation, proper security and clear processes must be implemented.

2.9 Research Gap

The information systems have evolved to include digital platforms, official alert dissemination, GIS, crowdsourcing and AI assistance for disaster management. However once information has been collected it must be organised, assessed, prioritised and linked to the response.

There are several reasons for developing BhimShila. Firstly, information from different sources must be easily accessible through one platform. Secondly, citizen reports must be organised to ensure they are processed correctly. Thirdly, volunteers and coordinators must have clear processes for handling missions, updates and monitoring incidents. Lastly, any AI assistance must have guidelines to ensure safety and human oversight. Processes must also be designed with field conditions in mind as connections may not be reliable.

BhimShila offers a solution to this by including external disaster information, citizen SOS reports, risk scoring, AI assisted incident management, duplicate report clustering, role based coordination and selective offline support.

This system will not replace official alert agencies, it also does not provide a new forecast model or complete coordination between all emergency organisations. Rather it aims to provide a platform which integrates select information sources with response processes.
Overall, the literature outlines a need for information integration, location awareness, structured reporting, responsible AI assistance and clear processes. This has been used to design BhimShila.

PROBLEM DEFINITION AND OBJECTIVES

3.1 Problem Definition

Disaster response teams often need a shared insight into hazards, locations, requests for help and response capacity. Often information is received from multiple sources which may not all be official or up to date. This can cause issues such as duplicated reports affecting the understanding of demand, partial reports delaying triage and unreliable connections causing issues with updates. Even if a system could collate messages, it may not be reliable if there is no explanation for why the alert is trustworthy, no oversight of who approved an action or confirmation that a dispatched team has responded.

BhimShila aims to provide a single, explainable and robust decision support system to link alert monitoring, citizen reporting and relief coordination whilst ensuring safety critical decisions are made by authorised humans.

3.2 Objectives

Collect official alerts with related environmental data from identified data sources.

Provide a regional dashboard to display official alerts, calculated risk and citizen SOS reports with traceability.

Convert emergency reports (verbal or typed) into structured data.

Ensure safety guidelines and prioritisation so that urgent reports are not de-prioritised by the language model.

Group likely duplicated reports based on category, distance and time.

Correlate select signals with supporting evidence. Hold low confidence signals.

Allow authorised coordinators to create alerts, dispatch teams, verify details and confirm/cancel actions.

Keep track of who owns the mission, changes in status, deployment activity and results from the field.

Hold and synchronise select reports/updates whilst connection is restored.

Test deterministic modules. Note limitations, dependencies and future validation.

3.3 Scope & Assumptions

The repository will give an overview of the nation however will provide more detailed regional insights with selected attributes. External services may not always be available or may be rate limited. Data provided by third parties may not always be of good quality. The seeded response units and demo data should not be assumed to be an accurate list of all response organisations. This is a prototype and will not replace emergency phone numbers, official alerts, local disaster management plans or trained responders.

3.4 User Roles and Responsibilities

Table 2: Function Roles

Role	Top Requirements	Features of Platform
Citizen	Quickly report an emergency, get relevant alerts and precautions	Voice/typed SOS; Local alerts inbox; Shelter lookup; Protocol lookup
Volunteer	Know your assigned work and report progress on limited connectivity	Task list; Status changes; Notes/photos; Offline queue; Directions
Coordinator	Review evidence, prioritise incidents, allocate funds, and approve warnings	Verification; Mission dispatch; Alert queue; Organization and resource management; Unified dashboard
Public authority/NGO	Keep an auditable response process	Deployment record; Operational summary; Action verification; Source of origin

PROPOSED METHODOLOGY AND WORK DESCRIPTION

4.1 Overall Approach

The system has a layered architecture which enables separate testing of components. The architecture separates the data acquisition, normalisation, analysis, incident management and co-ordination functionality so that they can be tested individually. The data adapters are responsible for gathering or reading data from the selected source. The normalisation component ensures the data is uniformly formatted. The analysis component calculates risk indicators and reconciles evidence. The incident co-ordination component provides this information to the correct role.

4.2 External Data Acquisition and Normalisation

NDMA SACHET/CAP is used as an upstream source to ingest official warnings. Weather and historical context are provided by Open-Meteo, earthquake event information is provided by USGS, river context is provided by GloFAS, selected geographic features are provided by OpenStreetMap and Overpass, and a set of news feeds provide additional reporting context. Each adapter is responsible for retrieving the data it needs and mapping the relevant fields to the application's internal representation.

The source types are intentionally different. A statement by an authority is official warning data. Forecast values are modelled environmental information. News is reporting second hand. A citizen SOS is an application filed by a user. A risk score generated by an application is a deterministic risk score. This allows the operator to assess evidence without assuming that all items are equally important, by presenting the categories separately.

Normalisation deals with timestamps, locations, categories, source names and status information where applicable. Repeated external retrieval may be unnecessary or unreliable and cached or bundled information may be used. The system should expose data age and availability of source, because a stale value should not be mistaken for a fresh observation.[6]-[9], [16]-[19]

Table 3: External Data Sources and Their Intended Use

Source	Information Used	Role in BhimShila	Important Constraint
NDMA SACHET / CAPS	Official warning messages and status fields.	Authoritative alert provenance and context.	Must check validity, message type, status, and availability of source.
Open-Meteo	Weather forecast and historical data.	Regional context – structured environmental input.	Forecasts are not a guarantee. API availability and resolution vary.
USGS Earthquake Event Catalogue	Earthquake event catalogue.	Seismic setting of the selected regions and time-windows.	Results are affected by catalogue coverage and magnitude/time filters.

GloFAS	River/flood-related model context.	Additional hydrological context.	Not a replacement for local gauges or official instructions.
OpenStreetMap / Overpass	Features such as waterways and facilities that have been mapped.	Nearby-feature and spatial enrichment queries.	Completeness and freshness are geography dependent.
Selected news feeds	Secondary coverage of developing stories.	Context to support coordinator review.	May be delayed, inaccurate, or irrelevant; not authoritative on its own.
Bundled district/shelter data	Centroids, district geometry, and shelter reference records.	Offline lookup (regional display and selected workflows).	Bundled data can go stale and should be versioned.

4.3 Regional and Geospatial Context

District boundaries and centroids enable rendering of regional maps and matching locations to regions. Selected shelter data is packaged to allow lookup when a live query is not available. Open geographic features can provide context, such as waterways and facilities. These datasets are to be considered as reference information and not as a complete and constantly verified inventory.

Where population or exposure characteristics are available, the platform can use these to provide context to the prioritisation process. The census values have a reference year and geographic resolution and should not be presented as current real-time population counts. The implementation must retain the source and vintage of such attributes so that the coordinator can understand the bounds of the estimate.[9],[10]

4.4 Deterministic Risk Assessment

Input data is normalised according to the expected input range for this implementation. The data is then weighted using explicit weighting and threshold rules. The output gives a result including a score/level along with an explanation of how each available indicator has contributed. This means we do not need to ask a language model to produce a risk score. This means we can easily replicate deterministic logic in our unit tests. It also gives us visibility into why a location has been prioritised. Whilst deterministic calculations will be used, this does not mean it is scientifically valid. We must still review the weights/thresholds with experts and compare them to historical incidents.

If there are missing data points we should let users know that the result is incomplete or reduced. Having a local score will help us to focus on what we need to but does not mean it will automatically tell us that there has been a disaster or an official warning has been given.[10][20][21]This design avoids asking a language model to generate a numerical risk value. Deterministic logic is easier to reproduce in unit tests, and allows to inspect why a location was ranked as higher priority. Deterministic calculation doesn't mean scientific validity; weights and thresholds still have to be reviewed by experts and evaluated against historical events.

If data is missing, then the system should indicate that the result is incomplete or degraded rather than silently treating missing inputs as safe. A local score helps organise attention, but does not, by itself, declare a disaster or issue an official warning.[10], [20], [21]

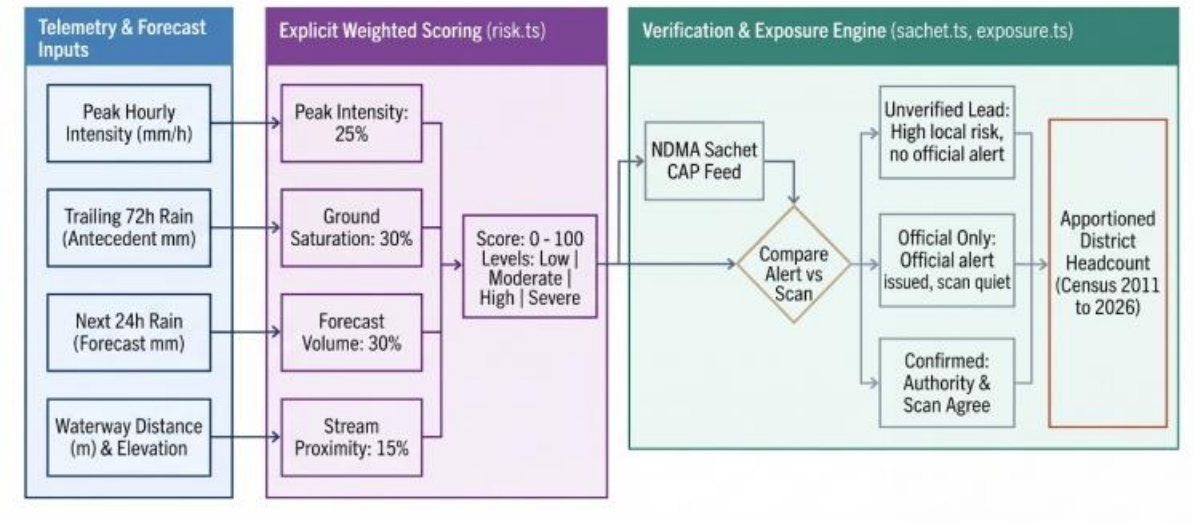


Figure 1: Risk Scoring & Ground Verification Engine

4.5 SOS Input & AI Parsing

Users will be able to input an emergency either through a text input or voice input (if supported by browser/device). The text will be passed onto the reporting layer. An AI parser will attempt to extract key information from the input. This will include brief summary, category, severity, number of people affected and location of the incident. The location of the incident can be identified using location data available or device co-ordinates if allowed. If there is no model service the system will handle this by generating a basic report from the text.

The models output is then considered as potential data. The fields needed will be validated and any unknown information will not be guessed. A deterministic validation will be performed to ensure critical keywords are included and that any high risk rescue reports cannot be reduced below the minimum severity level set. This will ensure that the model does not exceed the defined limits. [11][13][14]

4.6 Clustering & Prioritising

Upon receiving a report the incident processor will compare it to valid recent incidents. The current baseline identifies potential duplicates using category, distance (12km) and time (within 12hrs). If there is a potential match then the report can be associated with the existing incident. However, it will continue to be a separate report.

The rules here have been kept simple to allow testing but depend on the type of hazard and location. Large floods could cause many reports across a large area, or there could be two house fires within a small radius. In future versions there will be bespoke thresholds for each type of hazard, confidence indicators and a override for the coordinator. Reports will be prioritised depending on severity and if teams have already been assigned. This will highlight what needs attention. [11][12]

4.7 Validation & Evidence Matching

The validation layer will compare chosen local signals and match them to any supporting evidence. This could include official warnings, news, earthquake data and local submissions. If there is an active official warning then this will be considered an official alert. If there is no matching evidence for a high local score this would not be an official warning. If there is a status or type field in the source message this will be used to identify if the message is actionable or if it is a test/cancel/expired.

Evidence does not prove something is true but can help with reviewing. News may be delayed, irrelevant or incorrect. If there is no matching news items this does not mean there was no incident. If there is no source available this is just an issue with data availability. The interface will display evidence and where it came from so an authorised person can decide to hold, ignore or take action.

4.8 Coordinator Review, Alert/Dispatch & Confirmation

The coordinators will review the details of the incident, evidence, severity, location and if it has been assigned. If an alert/dispatch is required then the interface will put it into a pending state and display the supporting information to the relevant recipients/teams. The coordinator will need to actively confirm or deny the action.

Where possible the original wording and language from the official warning will be used. Actions taken externally will be idempotent and traceable. Using a confirmation gate will reduce the chances of accidentally broadcasting but staff will still need training.

4.9 Volunteer Management & Mission Notes

Volunteers will be able to view what they need to do for their organisation. They can confirm they have accepted the task, when they arrived and provide any updates or results from the field. The status changes will act as a lightweight log to track how far the volunteer has got with deploying. Mission notes will allow the user to add observations related to the task. Checks are in place to ensure volunteers can only view/edit certain records based on their role and organisation. Just because they can edit a mission does not mean they should be able to send public alerts. The coordinator is responsible for assigning tasks and any external designated actions. In future versions there could be improved resource modelling but for now the prototype should not be described as having verified inventories or inter-agency dispatching.

4.10 Offline Functionality

When there is no internet connection selected actions will be stored locally. When connectivity is regained the system will attempt to sync the actions with the server. Each action will have a unique ID to stop duplication. The system will show if the action is pending, synced, failed or stale.

Only certain functions will have offline functionality. Any old data will be marked as stale. Any pending SOS will be clearly shown as not having been sent to the response team.[15]

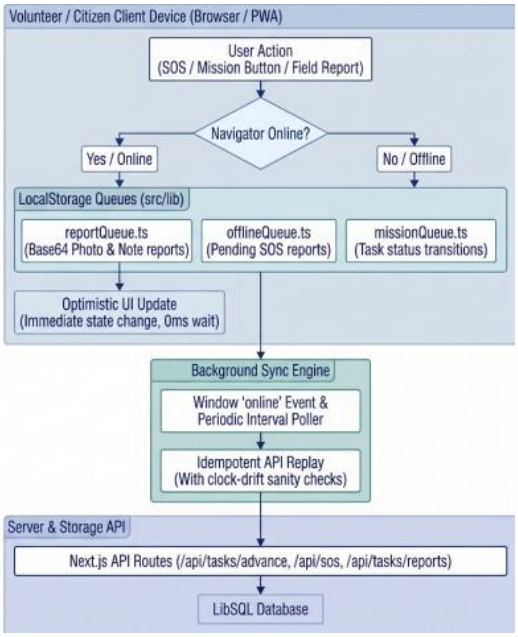


Figure 2: Offline-First Edge Resilience & Sync Pipeline

4.11 Security, Privacy, and Auditability

These notes cover how passwords are stored securely with scrypt, server side sessions, http-only cookies and role/organisation based access controls. There are ways to bypass controls on the client end therefore it is important to ensure validation occurs on the client. The data model should only allow access to personal data when required to answer questions and hold people to account.

A review of privacy and security within the system needs to be documented. This should include details of how location and contact information will be collected and stored, logging of access, secrets, backups, account cancellation, avoiding abuse and how data incidents will be handled. Just because a control has been mentioned in the code does not mean that the application is secure. [22]-[24]

4.12 Work Description

1. Requirements Analysis – Determine what roles are necessary, the process for reporting emergencies, data sources and prototype limits.
2. Data Integration – Create adapters for official alerts, weather, earthquakes, rivers, news and geographical data.
3. Intelligence – Deterministic risk scoring, incident parsing, safety floor rules, grouping duplicates.
4. Coordination – Connect incident review to assignment, confirmation, volunteer updates and records.
5. Resilience – Adding local queues for certain actions and sync behaviour for reconnections.
6. Validation – Run deterministic tests and certain integration/live source tests. Record any failures and assumptions.
7. Reporting – Map the features to evidence, note any limitations and what needs to be done before it can be deployed.

PROPOSED ALGORITHMS

The system integrates deterministic numerical, geospatial, rule based (NLP) and information extraction (AI) processes. Explicit rules are used to calculate numerical scores, thresholds, ranks and priorities. The language model helps aid understanding of the text, however cannot independently produce the numerical risk scores or override official warnings. Below are the algorithms used within the project.

5.1 Flash-Flood Exposure Scoring Algorithm:

Weighted Linear Scoring Model

The algorithm calculates a score between 0 and 100. The algorithm uses antecedent rainfall, forecast rainfall, peak hourly intensity and nearness to watercourse. Normalisations are applied to each of these variables (between 0 and 1), then multiplied by fixed weights. [20][21]

Formula

SCORE = 30 × antecedent_ratio + 30 × forecast_ratio + 25 × intensity_ratio + 15 × proximity_ratio

Table 4: flash flood exposure scoring

Factor	Weight	Normalisation
Ground saturation (rainfall over past 72 hours)	30	min(rain_72h / 150, 1)
Forecast rainfall (next 24 hours)	30	min(rain_24h / 100, 1)
Peak hourly rainfall intensity	25	min(peak_mm_h / 20, 1)
Watercourse proximity	15	Linear decay: 1 at ≤100 m; 0 at ≥1,500 m

Table 5: Risk Level

Score	Level
0-27	Low
28-49	Moderate
50-69	High
70-100	Severe

Scores of 50 or above are flagged as higher than the model. This indicates priority, not a precise likelihood of flooding.

5.2 Prioritising alerts based on population at risk

Algorithm: Two Tier Comparator Sort

Official SACHET alerts are sorted by IMD colours (Red = 3, Orange = 2, Yellow = 1, Green = 0) and population at risk.

Comparator

```
sort(a, b) = COLOUR_RANK[b] - COLOUR_RANK[a] || b.people - a.people
```

The population gives a hierarchy to the alerts based on severity. This means that an alert with a high population estimate will never override an alert with a greater severity.

5.3 Estimate Population Exposure

Algorithm: Named District Search + Geometric Circle Intersection

If the alert mentions specific districts then a search of the normalised lookup table will be The following will be performed. Where multiple instances of one district occur, help from the issuing state will be provided. Where no districts are specified, the alert region and districts will be approximated as circular. [25]

```
Estimated population contribution = total_population × (overlap/district_area).
```

Where there is less than 2% overlap in the data, it will be rejected. The population will be calculated using data from the 2011 census and district growth rates to predict the population for 2026.

5.4 Rainfall anomaly detection

Algorithm: Historical Percentile Ranking

For a particular location:

1. Get the complete time series of precipitation for that day from Open-Meteo ERA5 reanalysis (~250kb 1940 to present).
2. Calculate the current 14-day accumulation of rain (currentMm).
3. For each previous year find the same calendar fortnight and accumulate the total rainfall.
4. Determine the percentile of currentMm against the historical distribution.

```
percentile = (years_with_less_rain / total_years) × 100
rank       = 1 + count(years_with_more_rain)
```

Table 6: Rainfall Anomaly Classification Based on Historical Percentile Rankings

Percentile	Verdict
≥99	Exceptional
≥95 and <99	Unusual
≥85 and <95	Elevated
≥30 and <85	Normal
<30	Dry

The percentile and verdict are deterministic. This gives an indication of how unusual something has historically been.

5.5 SOS Triage and Classification

Algorithm: Keyword Parser → LLM Extraction → Deterministic.
Two stage algorithm:
Keyword Heuristic Parser
This takes free text in Hinglish/Hindi/English and uses keyword matching to identify which category it should fall into. It can interpret words like phans/phase/phanse/fanse (फंसे) which means trapped and doob/dub (डूब) which means drowning. It also determines that if someone says paani ghus gaya this means water coming up and therefore pine ka paani (water rescue) and will return food_water. This returns:
{
category,
severity (1-5),
peopleCount,
place,
summary
}
LLM Extraction
An LLM (4 seconds) is used to extract structured JSON from the message to allow for better data (named places, number of people and summary in clean English).
Reconciliation
The keyword parser acts as a minimum.

```
final.category = keyword_is_life_safety ? keyword.category : model.category
final.severity  = MAX(keyword.severity, model.severity)
final.place     = model.place ?? keyword.place
final.summary   = model.summary || keyword.summary
```

The LLM would have labelled this incident as severity 4, under the category Shelter due to the context of the quote "paani ghar me ghus gaya, chhat pe phase hain" which describes individuals stuck on a roof due to rising water. However, the keyword parser would correctly label this as severity 5, under the category Rescue. [11] [13] [14]
Categories: Rescue, Medical, Food/Water, Shelter, Missing, Other
Severity: 1 (Information) up to 5 (Life in immediate danger).

5.6 Incident Clustering

Algorithm: Spatial Temporal Cluster.
Parameters
Cluster Radius: 12km (CLUSTER_RADIUS_KM)
Cluster Window: 12 hours (CLUSTER_WINDOW_HOURS)
Upon receiving a new SOS:

1. Check all active incidents of the same category from the last 12 hours.
2. Calculate the Haversine distance between the new SOS and existing incidents.
3. If there are any incidents within a 12km radius:
Merge them together. Increment report_count, Set severity to MAX(severity, new severity), Set number_of_people to MAX(number_of_people, new number of people).
4. Else Create new incident.
This will allow an incident to be created where if there are 10 calls regarding a single bridge collapse, only a single incident will be created, but will have 10 reports saying the same thing. This directly solves the problem of duplicated calls after a disaster such as the Aceh and Haiti earthquakes.

5.7 Incident Prioritiy

Algorithm: Weighted Sum with Anti-Starvation and Coverage Penalty.

```
Priority = severity_score + corroboration_score + people_score + waiting_score + coverage_penalty
```

Table 7: Components and Weighted Factors Used in the Incident Prioritisation Algorithm

Component	Formula	Maximum / effect
Severity	severity × 14	70
Corroboration	min(20, (reportCount − 1) × 7)	20
People affected	min(18, $\log_{10}$(peopleCount + 1) × 14)	18
Waiting time	min(18, (waitingMinutes / 60) × 6)	18
Already assigned	−30	Penalty

Anti-starvation: There is a waiting time, meaning incidents will always be addressed. If you have an 18 point bonus for waiting 3 hours. This can be better than a low severity incident.
Penalty: When an incident is assigned, you get -30 points. This means more responders will be brought up for unassigned incidents.

5.8 Deterministic Intent Router

This uses a Rule Based NLP Router. It utilises keyword matching and regex extraction. This replaces the LLM based router, which used ~1470 tokens of schema per turn. This has no cost and is very quick. It supports Hinglish, Hindi (transliterated) and English.
The router contains:
Date Extractor using regex. Supports ISO format (2025-08-16) and Natural Language format (26 June 2025).
Place Extractor using regex. First detects labelled districts (Indore district). Then detects places with a preposition attached (risk in Kerala). Finally detects genitive places in Hindi (Kangra ka risk). It also checks if it is a real place.

Command Filter detects whether the command is a send or broadcast command. This prevents confusion between the two, as if confused a team will be sent when a broadcast was required.
It handles the following commands:
situation_report
past_weather
dispatch_team
list_incidents
official_alerts
prepare_alert
find_shelter
lookup_protocol
verify_signal
search_news
list_hazards
assess_risk
rank_regions
region_profile
greet
If the query is ambiguous, it will return null. This is decided by the LLM.

Summary of Algorithms

Table 8: Summary of Algorithms

S. No.	Algorithm	Purpose
1	Weighted linear scoring	Flash-flood exposure score (0–100)
2	Two-tier comparator sort	Alert ranking by colour severity then population
3	Named lookup + geometric disc overlap	Population exposure estimation
4	Historical percentile ranking	Rainfall anomaly detection
5	Keyword floor + LLM reconciliation	SOS classification with safety floor
6	Spatial-temporal clustering	Associate likely duplicate SOS reports
7	Weighted additive priority score	Prioritise incidents
8	Rule-based NLP router	Route clear user intents

PROPOSED FLOWCHARTS

6.1 System Architecture

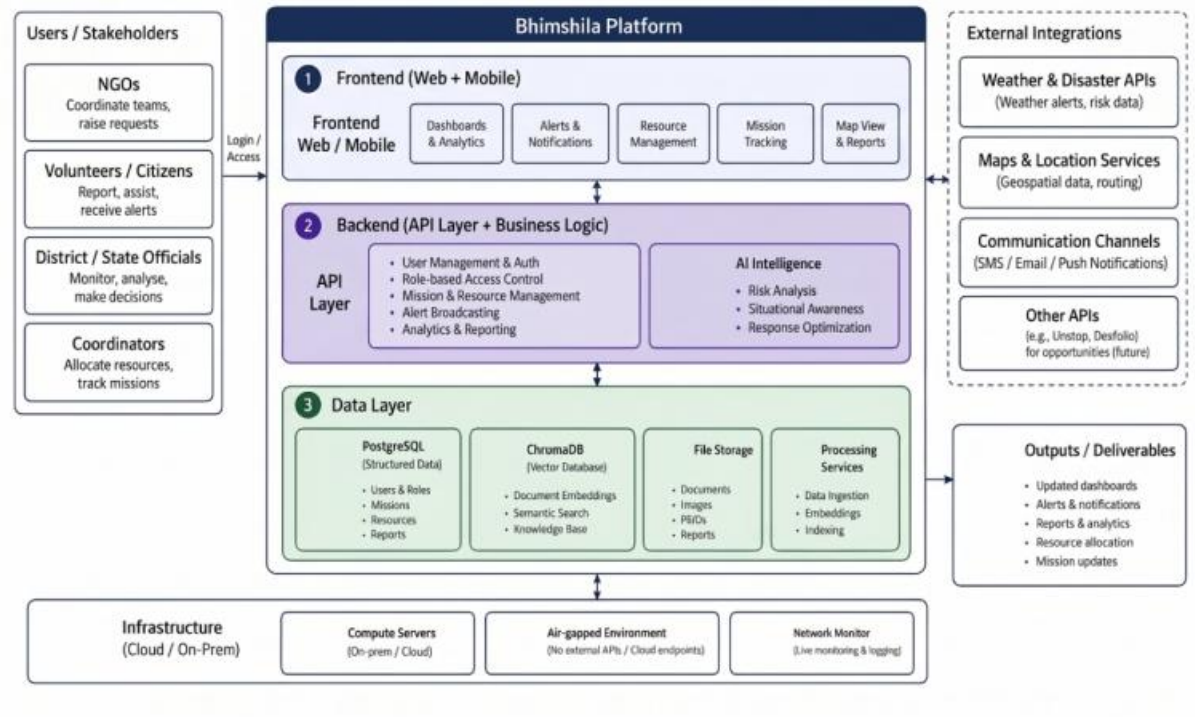


Figure 3: System Architecture

End to End System Workflow

The system starts . Users go to the platform through web or mobile interfaces . Depending on their assigned roles, users can file incident reports, request resources, get alerts and monitor active missions. The frontend sends requests to the backend through APIs, which are authenticated and processed according to the business logic of the system.

The backend pulls/updates relevant information in the data layer and uses external integrations as and when required. The AI intelligence module reviews the information available to support risk assessment, situational awareness and response prioritisation. The system then generates relevant notifications, analytical reports, dashboard updates and resource allocation recommendations from the processed information.

Coordinators and authorised officials can review generated information and make operational decisions . This ensures that human supervision is maintained over critical response activities . Then, any generated updates are communicated to the concerned stakeholders via the platform’s available communication channels.

6.2 Database Design and Entity Relationship Model

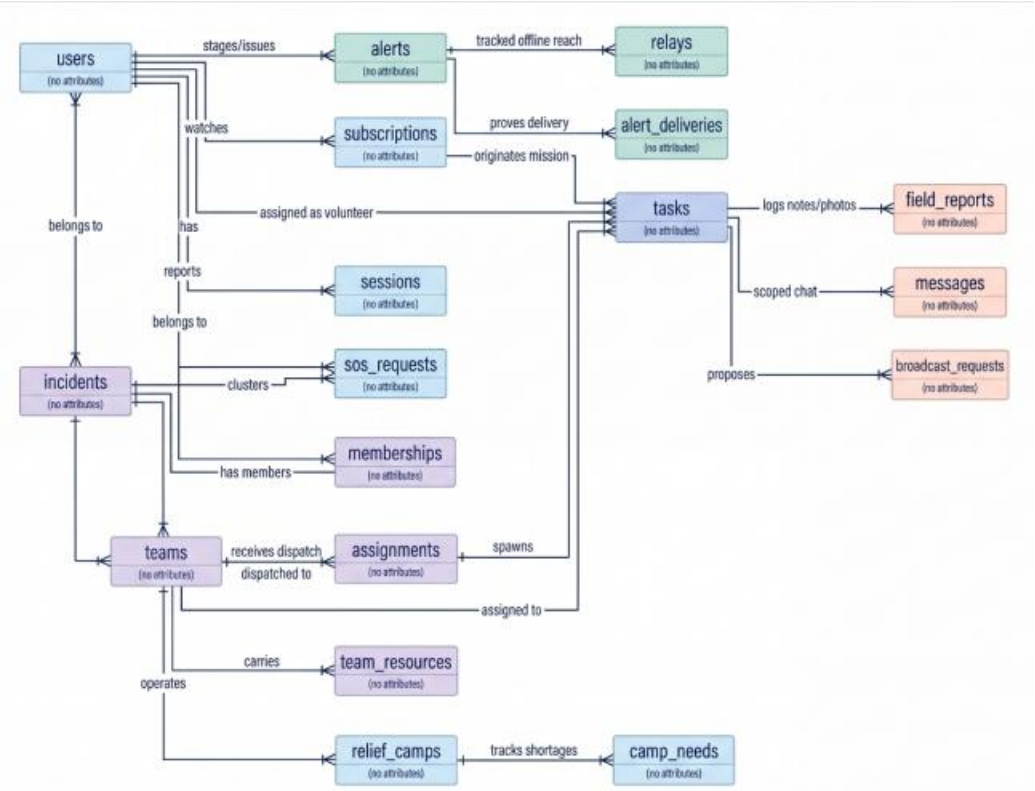


Figure 4: Relational Database Schema & Data Model

Key Considerations and Invariants

There are various constraints within the database to ensure consistency, security and traceability throughout the database.

1. Separating roles: This allows us to separate permissions given to users at platform level and at organisation level. Users have a role within the platform which can be found using the column `users.role`, this can be a citizen, volunteer or coordinator. Users have a role within an organisation which can be found using the column `memberships.role_in_org`, this can be lead or member.
2. Single source of truth for organisations: The `teams` entity is used to represent an organisation and removes the need for different organisational structures for each response group. This makes it simpler to manage who is on a team and what resources they have.
3. Source traceability for a task: A task will always have a source. This can either be an incident reported by a citizen using the column `incident_id` or an alert using the column `alert_id`. This ensures we can track any information that came into the system relating to a disaster and how it has been responded to.
4. Duplicated alerts: Using the `seen_alerts` functionality allows us to store the alert identifiers that have been processed. This will then be checked when the system does its regular hourly synchronisation so duplicate notifications aren't created. This makes the notifications more reliable and ensures a consistent history of alerts that have been processed.

6.3 Volunteer Task Allocation and Field Response Workflow

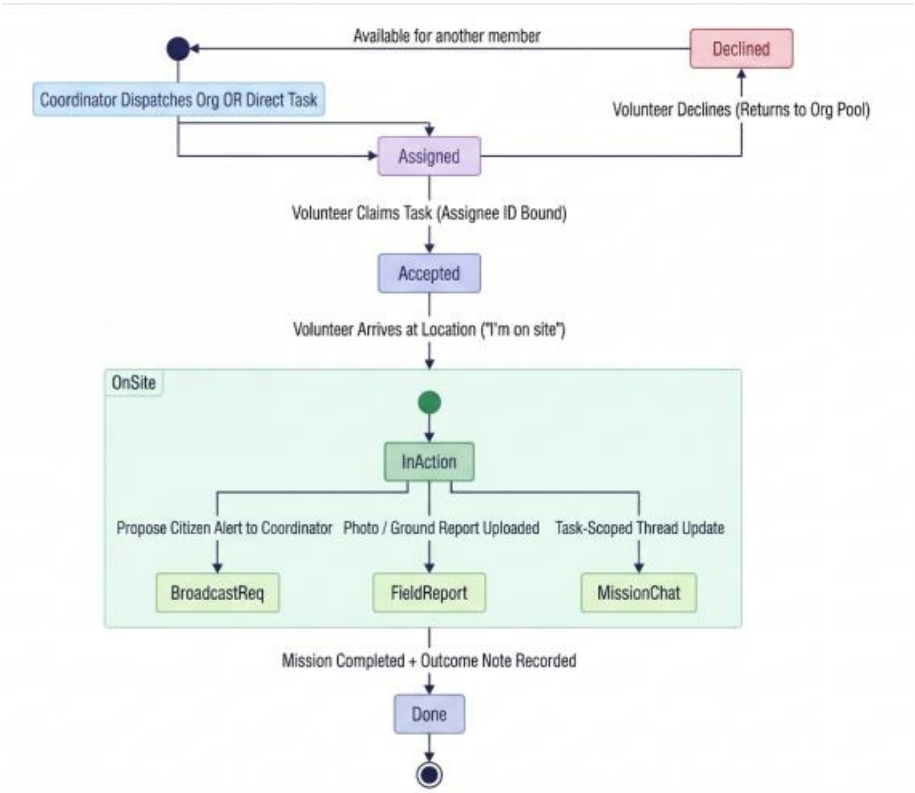


Figure 5: Field Operations, Missions & Deployment Log

Key Technical Components

1. Activity Logging

The platform logs activities using timestamps. These timestamps are logged in fields such as `accepted-at`, `on-site-at` and `done-at`. If there is no internet connection, updates can still take place by logging locally and syncing with the main backend. This makes it so that activities in the field aren't affected by internet.

2. Alert Broadcasting

Volunteers cannot broadcast official warnings, however they can suggest alert messages. These suggestions are logged as `broadcast_requests` and sent to an authorised coordinator. Once authorised the alert can be broadcasted to appropriate channels such as Telegram and in-app alerts.

3. Out of Platform Communication & Relays

The platform can monitor alerts that are sent outside of the platform. Volunteers can log the number of households they have reached using the `relays` feature. They can also log how they communicated with the households e.g. face-to-face, phone call, public announcement. This helps to ensure that the warnings have been passed onto the community.

4. Relief Camp Matching

This matching feature links the shortage of resources reported by relief camps with the available resources of the response teams. It identifies the need for key resources such as food, drink, medication, sleeping equipment, sanitary products and rescue boats and compares this to what the teams have in stock. This allows teams to prioritise resources and for coordinators to identify any gaps in resources from multiple relief camps.

IMPLEMENTATION (TOOLS AND TECHNOLOGY USED)

7.1 Technology Stack Overview

It's a web application with typed front end, server-side API routes, persistent data storage, and adapters for external services. The stack chosen is designed for responsive user interfaces and modular service logic. The problem is that the dependencies are not specified in the package lockfile.

Table 9: Core technology stack and rationale

Technology / service	Role in the project	Implementation rationale / caution
Next.js	Application framework and routing.	Supports page rendering and server-side route handlers.
React	Component-based user interface.	Reusable role-aware views and interactive operations screens.
TypeScript	Application logic and data models.	Static typing supports safer changes across adapters and workflows.
Tailwind CSS	Interface styling.	Responsive layout and consistent visual treatment.
LibSQL / SQLite-compatible database	Persistent application records.	Stores relevant user, incident, mission, and workflow state; deployment configuration matters.
NDMA SACHET / CAP	Official warning source.	Preserve issuer, status, validity, and original instructions where available.
Open-Meteo	Weather forecast and historical context.	External dependency; output is input to prioritisation, not a guarantee.
USGS Earthquake API	Earthquake event context.	Provides catalogued event data subject to source coverage and update time.
GloFAS	River / flood context.	Hydrological context is not a substitute for local gauge or authority information.
OpenStreetMap / Overpass	Geospatial features.	Feature completeness and freshness vary by region.
Selected news feeds	Corroborating context.	News is secondary evidence and requires source/time checks.
Automated test scripts / GitHub Actions	Regression and scheduled checks where configured.	CI configuration is not evidence of successful production monitoring unless verified.

7.2 Front End

Front End is used to create a role-specific view for the citizen, volunteer, and coordinator. This has been built using best practices from Next.js [18], React [19], TypeScript [20], and Tailwind CSS [21]. This gives a consistent view for users to see incident statuses and the evidence required. A responsive approach should be taken to ensure the site works well on mobile devices for the citizen and volunteer. Whilst the coordinator may be on a larger screen.

The reporting tool has the ability for text and speech inputs where supported by the browser. Speech to text accuracy will vary depending on device, language and environment, so text input is important to have. If issues occur whilst trying to convert speech to text then the report should not be lost. This should be indicated if the user cannot gain permission to use their microphone.

The coordinator view combines alerts, risk context, incident records, and mission statuses. The labels or evidence should be clear to show what is an official alert, what is risk inferred locally, what is a report from the user, and what is cached. This avoids giving the impression all items have the same status.

7.3 API and Service

The API routes form the backend edge of the application. They take request data and validate it, enforce role and organisation boundaries, and update the state with allowed changes. Adapters have been created to separate external sources from the application. This helps reduce coupling and means individual adapters can be tested if the provider changes schema or goes offline.

Validation on the backend is important for processes like assigning missions, changing task status, or confirming alerts. It would not be secure to hide buttons on the frontend. The server must always know who is requesting a protected action and what their permissions are.

7.4 Data Persistence & Incident Life Cycle

Data persistence handles storing data required for reporting, incident grouping, missions, user roles, and action statuses. We've chosen SQLite compatible options [22]. Reports and incidents are separate but related. There may be multiple reports to provide evidence for one incident, so original reports should still be accessible. Mission statuses are logged to show who has been assigned and what has happened in the field.

To reduce duplication in the client when re-trying a request, unique IDs should be used and validation performed on the server. Local queue is used for temporarily storing work to be performed offline. Authoritative data is data from the server once it has been validated and stored. Conflict resolution and partial failure should be tested.

7.5 External Integration

NDMA SACHET/CAP is providing the information for warnings. Handling the status and type of message is important as there may be updates, cancellations, tests, or expired alerts. If full CAP documents are available then the structured data and instructions from the issuer should be kept. [5]-[7]

Open-Meteo provides weather context, USGS provides earthquake records, GloFAS provides river context and OpenStreetMap/Overpass provides selected geographical features. News is

provided for secondary reporting. All adapters should handle timeout, malformed data, rate limiting, missing fields and unavailable sources. In the event of a failure we want to see a degraded state.
Reference data may be bundled to allow selected features to be used without calling live data. This improves availability but should have a responsibility to keep the data fresh. The build should note where the data set has come from and what date the version was. The interface should not give the impression that this data is being updated regularly. [9] [16]-[19]

7.6 AI Parsing & Falling Back

AI parsing helps parse the unstructured descriptions of incidents and interpret selected commands. This means there is no tight coupling between the natural language processing and the deterministic scoring of the environment. Fields suggested by the model are validated before being added to the operational data. Safety floor is used to cap the severity of critical phrases.
There are several reasons why the model service may fail. Connectivity, rate limiting, changes to the provider, and invalid output. Some report processing has falling back. Falling back should be considered using example Hindi, English, and Hinglish. We do not claim that the parser does not hallucinate or find all critical incidents.

7.7 Local Queue & PWA

The project has local queuing for selected SOS and mission actions. Once the network is restored any queued items will be synced. This helps to maintain continuity in the field. Unique IDs, retries, validation, and an error state should be used. Offline functionality should be tested to consider intermittent connectivity, page refreshes, expired sessions and duplicate retries.
Having an installable or PWA style application can help with accessibility on mobile. However having an application installed does not guarantee that all routes or data sources are available offline. Live alerts and weather rely on the upstream source unless there is a recent cache. The interface should show how old the cached data is. [15]

7.8 Authentication & Authorization

Password hashing using scrypt, Sessions on the server, Http Only Cookies, and Role Based or Organisation specific checks have been implemented in the repository. This provides a baseline to help restrict access to user and mission data. Any sensitive actions should be validated through the server.
Before deploying to production the application should be reviewed for security. Sessions will expire, Brute force protection, CSRF protection (if applicable), Secret storage, Database access, Dependency vulnerability, Logging, Backups, and account recovery. Emergency workflows should also be considered. The public SOS endpoint could be abused so should have rate limiting and moderation. [22]-[24]
7.9 Testing & Development
Deterministic tests have been added to the repo. Some live checks have been separated. Once the workflow has been turned on and setup checks can be automated using GitHub Actions [23]. Separating live checks from deterministic tests helps to avoid confusion between

temporary outages from the provider and issues in the core logic. It also helps stop any false confidence in the fixed fixtures.
There is automated test coverage for multiple test suites and hundreds of assertions. The exact number varies depending on the note in the repository. The final submission should have the count from running a new test on the last commit. The output from the test should be attached or saved. We do not indicate that the tests were re-run.

7.10 Deployment

The application relies on the database being configured, environment variables, external APIs and hosting. In production it should have secure secrets, a managed backup and restore process, monitoring for failed adapters, and a documented upgrade process. Run your database migrations using a copy of realistic data before shipping.
For an operational deployment you should have an agreement on who can create accounts, any organisations represented, what alerts can be sent, who oversees the coordinator role, and how to handle a system outage. The system will not be usable in an emergency if it is not technically available. There should be a human model and clear escalation paths.

RESULT DISCUSSION AND ANALYSIS

8.1 Evaluation

There are three key points which need to be considered when evaluating the software: Does it perform its intended functions? Does it integrate well with upstream data sources? Does it provide an improvement in the real world? This can be answered in terms of the first two by automated testing. However, the last point requires field studies, expert review, realistic scenarios and outcome measures to be carried out. This cannot just be tested from code.

The findings presented here have been taken from the testing processes carried out. This is not a new field trial, controlled trial with emergency response teams or a statistical calibration study. Therefore, these findings only show that it has been implemented and validated, but do not prove it can predict or operate.

8.2 Test Coverage

A table outlining the main areas of testing and evaluation carried out can be seen below (Table 10). New testing logs or screenshots from the final commit should be included in the final project team.

Table 10: Test and evaluation coverage

Area	Evidence available in project documentation	What it supports	What it does not establish
Deterministic logic	Repository contains test scripts for selected core logic.	Repeatable checks for defined input/output cases.	Completeness for all real-world cases or predictive accuracy.
Report parsing / triage	Parsing and safety-floor rules are covered by project tests.	Expected behaviour for tested phrases and edge cases.	Recognition of every language, accent, noisy transcript, or emergency.
Source adapters	Selected integration and live-source checks are documented.	Compatibility with responses observed during the check.	Continuous upstream availability or future schema stability.
Incident clustering	Configured category, distance, and time rules are implemented.	Deterministic grouping for tested records.	Correct grouping for every hazard type or geography.
Offline queue	Selected report and mission actions can be queued.	Continuity of supported actions under tested conditions.	Complete offline operation or guaranteed delivery without reconnection.
Role / confirmation workflows	Role-scoped operations and confirmation steps are part of the design.	Separation of selected responsibilities when tested.	Formal security certification or resistance to every abuse scenario.
Field effectiveness	No field evaluation is claimed in this report.	Not applicable.	Reduced response time, lives saved, or verified real-world accuracy.

8.3 Functional Outcomes

The workflow brings together several aspects of the project: external alerts, monitoring regional risk, citizen reporting, clustering incidents, coordinator review, assigning missions and updating volunteers. The most significant aspect of this workflow is that the system has a shared interface between them. This allows a workflow to be established from receiving signals/SOS reports to tracking the action taken.

Using the deterministic risk layer gives more insight into why a score is given, compared to using free text. The score will always be based on the inputted variables and weights. However, until the scoring model has been calibrated and tested against suitable historical data and labelled by experts it should not be referred to as a prediction model.

The safety floor rule ensures that any report with a critical rescue factor set will not be downgraded by the model. This is a targeted way to ensure reports are not downgraded. It is possible to have false negatives where the words used do not match those set out in the rules. There is also a possibility of false positives if the phrase is used hypothetically or incorrectly.

The duplicate association rule helps to avoid duplicate work. If two reports have the same category and are within the specified time and space they will be associated. However, this could cause incorrect matches. The original reports should be stored and evidence for why the reports were associated should be apparent.

8.4 Trustworthiness of Sources and Data

Integrating external data can increase the variety of data but can cause issues with reliance. You may not be able to receive an official alert feed, you may receive incomplete weather forecasts from the API, facilities may not appear in the map query or the news report may not mention the area of interest. A strong interface is required to show the source of the data and how up-to-date it is. You should not get confused between a source not being available and there being no danger.

There are multiple sources of evidence in the system; official alerts, local risk scores, secondary reports and citizen reports. The system has been designed so that this distinction remains clear. The coordinator needs to know what they know, where it came from, when it was updated and what they don't know. This is particularly important if there are two conflicting sources. [5]-[9][16]-[19]

8.5 Offline Functionality

The offline queue increases the chances that the selected reports or mission changes are stored when the system is offline. However, there are new ways that it can fail. The user may close the browser, the session may timeout, the server may not accept old mission changes or the client may attempt to perform an action again. To ensure reliable functionality there should be consistent identifiers, validation on the server, idempotent actions and visibility of queued actions.

This should be tested using consistent scenarios. For example, disconnect before submission, disconnect after server accepts but before receiving a response, reconnect after a long delay, reattempt the same entry, attempt to change missions with conflicting changes on two devices. This should not be seen as a guarantee of reliability in the field but an explanation of the functionality of the queue. There are still validation requirements that need to be met. Until the scenarios are documented and passed. [15]

8.6 Comparative Analysis

Table 11: Comparative view of related approaches and BhimShila

Dimension	Alert portal / feed	Single-purpose reporting tool	BhimShila prototype
Primary function	Publish official warnings.	Collect reports or requests for help.	Connect monitoring, report intake, evidence review, and coordination.
Environmental context	Provided by issuing authority or source.	Often limited to the submitted report.	Combines selected official, weather, river, seismic, and geographic inputs.
Incident organisation	May not track local duplicate requests or missions.	May organise reports but not upstream alerts.	Uses a configured category/distance/time rule and retains report records.
AI-assisted intake	Not necessarily part of the alert feed.	Depends on the specific tool.	Uses language processing with deterministic safety constraints.
Human approval	Authority process controls official alerts.	Varies by tool.	Requires confirmation for designated outward-facing actions.
Offline behaviour	Depends on provider/application.	Depends on implementation.	Queues selected SOS and mission actions; not all data is offline.
Evidence of effectiveness	Authority-specific operational evidence.	Tool-specific evaluation.	Software tests documented; field effectiveness not yet established.

This comparison is about functional emphasis, not about the superiority of BhimShila to the official systems. It isn't meant to replace any alerting and response capabilities you have in place, but to connect your selected workflows. Any claims about speed, accuracy or outcomes would need to be assessed directly and appropriately.

8.7 Limitations

- Risk scoring is based on configured deterministic weights and is not calibrated to a probability of a disaster.
- Map data and external services may be slow, incomplete, rate limited or unavailable.
- AI-assisted parsing may be misled by informal or noisy reports; safety-floor rules only cover configured indicators.
- Duplicate thresholds can over-cluster different incidents or under-cluster reports of a single widespread event.
- Offline support is selective: queued actions are not delivered until synchronisation succeeds .
- Full operational integrations and cross-agency coordination are not established with the current prototype.

- Active status will not be assigned to notification channels that need provider credentials or approval until they have been verified.
- This report is not intended to be a field study, formal security audit, or calibrated predictive evaluation.

Table 12: Current limitations and recommended future work

Current limitation	Risk / consequence	Recommended next step
Risk score not calibrated	Score could be misread as probability or forecast.	Validate against historical events and domain-expert labels.
External source dependency	Feed outages or schema changes may produce stale or incomplete context.	Monitor freshness, test failure modes, and show degraded states.
AI parsing uncertainty	Informal or noisy reports may be misclassified.	Evaluate multilingual samples and maintain human review.
Fixed duplicate thresholds	Separate incidents may merge or related reports may remain separate.	Tune hazard-specific thresholds and measure both error types.
Selective offline queue	Queued actions are not delivered until synchronisation.	Test retries, restarts, conflicts, and explicit delivery receipts.
No field trial claimed	Usability and operational benefit remain unmeasured.	Run supervised exercises with responders and community representatives.
No formal security audit claimed	Prototype controls may not cover all threats.	Perform threat modelling, privacy review, and independent security testing.

8.8 Recommended Evaluation Plan

Build a dataset of labelled synthetic and historical incident descriptions. This should include multiple categories, languages and severity levels.

Document historical incidents, risk scores (including false positives/negatives and uncertainty) and domain expert feedback.

Test incident clustering within urban, rural, flood, earthquake and localised emergency scenarios. Track any over clustering or under-clustering.

Test source adapters with fixed fixtures and live endpoints. Capture the time stamp, schema, latency and fault tolerance.

Test offline queuing with variable connectivity, browser refreshes, session timeout,

duplicate requests and conflicts.

Test with usability testing with citizens, volunteers and coordinators. Simulate safely and track how long it takes to complete, error rates and if users understand the status.

Conduct a security and privacy audit before using with real personal data.

CONCLUSION AND FUTURE SCOPE

9.1 Conclusion

The BhimShila is a web based prototype which has connected together various selected disasters monitoring sources, allowing SOS messages to be reported by citizens, incidents processed, reviewed by a coordinator and coordinated for relief. The main aim was to reduce the disconnect between the original signal and the action being taken to address the issue. The system has brought all sources of official alerts, environmental data, geographical and user submitted reports, but keeps the distinction between alerts and assumed information.

The project brings together deterministic risk scoring and AI assisted natural language processing. The deterministic risk provides insight into why a risk has been assigned and the natural language processing helps to make sense of unstructured reports. The safety floor reduces how much the risk of a critical report can be reduced and categorisation, distance and time rules mean similar reports can be linked together. The coordinator will need to verify the actions that are going out. Some functions can be placed in the queue to be executed when connectivity is available.

This is an example of an integration focused disaster information system. However, this has not demonstrated how accurate it is at predicting in the real world through software development and automated testing. The risk scores need to be tuned, the AI assisted intake needs to be further tested, external data needs to be kept fresh and the offline functionality needs to be tested with different scenarios. This should be considered as an ongoing decision support tool and not a replacement for official alert systems or trained emergency responders.

9.2 Future Scope

9.2.1 Risk model tuning and uncertainty

The deterministic risk model could be evaluated using historical weather, flood, earthquake and incidents along with expert feedback. Future work should document feature definitions, weights, missing data and tuning. The risk scores should also include uncertainty and should not be called probabilities until a valid statistical model can be used.

9.2.2 Incident Clustering

The current incident clustering rule is a basic 12km/12hr threshold. In the future this can use hazard specific rules to cluster incidents. This can include thresholds, confidence scores, similarity and domain expert feedback. The results should be evaluated to see if there are any false positives or false negatives.

9.2.3 Multilingual and accessibility

Test the report parsing with Hindi, English, Hinglish, regional words, poor audio quality and low bandwidth. Future improvements could be allowing users to select a particular language, verifying the extracted location names, screen reader support and alternative input methods. It should also allow users to type to report.

9.2.4 Offline reliability

The offline queue could be improved with persistent storage, idempotent keys, conflict resolution, expiry rules and confirmation of message delivery. A test harness should be created to test with unreliable connections and device reboots. It should also be possible to tell if a message was saved locally or received from the server and seen by the coordinator.

9.2.5 Integration

In the future this could be integrated using verified SMS or messaging services. These channels will need credentials, approval from the provider, consent, privacy considerations, rate limiting and monitoring. There will also need to be agreements on roles, data sharing, escalation and responsibilities between agencies. Any integrations should not be marked as active until they have been tested end to end in the target environment.

9.2.6 Security, privacy and auditing

A threat model, access control review, privacy assessment, secure deployment checklist, backup and restore test and incident response plan should be performed before putting this into operation. Audit logs should show what actions were taken as a result, however personal data shouldn't be stored unnecessarily. Retention rules and account deactivation processes should be agreed with the responsible organisation.

9.2.7 Field trials and human factors

The most important next step is a supervised pilot with practitioners of disaster management and representatives of the community using simulated or approved exercises. The study should capture user understanding of evidence labels, the extent to which the interface supports users in identifying unmet needs, the utility of duplicate grouping, and the clarity of confirmation gates. No live public warning shall be issued as part of an experiment without proper authorisation.

9.3 Final Remarks

BhimShila contributes with a common prototype for monitoring, reporting, triage and coordination with explicit safety boundaries. The future of this will be in careful validation, trustworthy data handling, operational partnerships and transparent communication of uncertainty. A good disaster management tool should help people make better informed decisions while keeping the authority and accountability with the people and institutions responsible for protecting the public.

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